

**CE Passed, RoCH, Reach certificated**

usage advance Demo: <https://youtu.be/3ZBea1PEKYM>

Pure Hardware setup demo: <https://youtu.be/ZUsadeHA6hM>

**KW007S/L Sensors**

*It's a simple, accurate, controllable, low power, motion sensor,  
sensed through non-metal, also work well together with BLE, Wifi.*

Full support directly from Keywave (linkedin / [x.com](#) / email)

**DIP Switch and detail Datasheet in Github:**

[https://github.com/namexavailable/Keywave\\_Modules/blob/main/KW007/KW007\\_Modules\\_EV\\_K\\_Guide.pdf](https://github.com/namexavailable/Keywave_Modules/blob/main/KW007/KW007_Modules_EV_K_Guide.pdf)

**Ai ready - Programming Guide:**

[https://github.com/namexavailable/Keywave\\_Modules/blob/main/KW007/KW007\\_Programming\\_Guide.pdf](https://github.com/namexavailable/Keywave_Modules/blob/main/KW007/KW007_Programming_Guide.pdf)

**Ai and maker ready - API and Code example plus explanation in comments:**

[https://github.com/namexavailable/Keywave\\_Modules/tree/main/KW007/code](https://github.com/namexavailable/Keywave_Modules/tree/main/KW007/code)

---

**Purchase Link in Digikey:**

<https://www.digikey.co.uk/en/products/detail/keywave-technology-limited/KWEVK0072SE2L2602/28960699>

Which model to Pick:

KW007-S in Digikey Range: up to 5 meters

KW007-L in Digikey Range: up to 12 meters

Or an Evk Kit for both sensors with a Evk board in digikey!

**Typical Applications**

KW007 is designed for reliable motion-triggered activation in environments where conventional PIR sensors may suffer from false triggering or environmental instability.

**Smart Building & Energy Applications**

- Smart lighting control, HVAC activation, Corridor and stairwell automation
- Energy-saving occupancy-based switching, Lighting inside enclosed fixtures

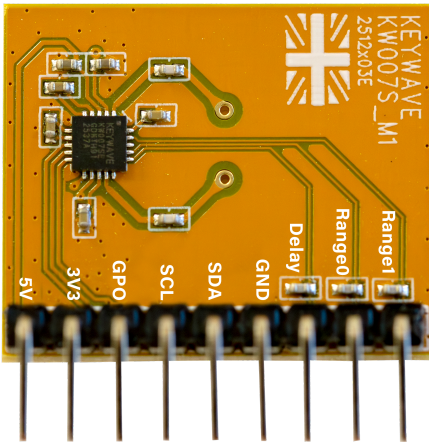
**Commercial & Public Installations**

- Interactive kiosks and displays, Digital signage wake-on-motion
- Automatic switches and panels, Equipment activation in public spaces
- Access control pre-trigger systems

**Industrial & Embedded Systems**

- Machine wake-on-motion, Enclosed equipment detection, Temperature-variable environments
- Vibration-prone installations, Compact embedded product designs

## Fast Setup (pure hardware):



This is the **KW007-SE 5.8 GHz** motion sensor.

It detects motion and outputs a digital pulse on the **GPO pin**. This signal can be directly connected to a relay or an MCU input to trigger your system.

### Minimum connection:

- VCC (3.3 V or 5 V)
- GND
- GPO

With this basic setup:

- Detection range: up to ~5 m
- Output: GPO goes HIGH (~1.9 V)

when motion is detected

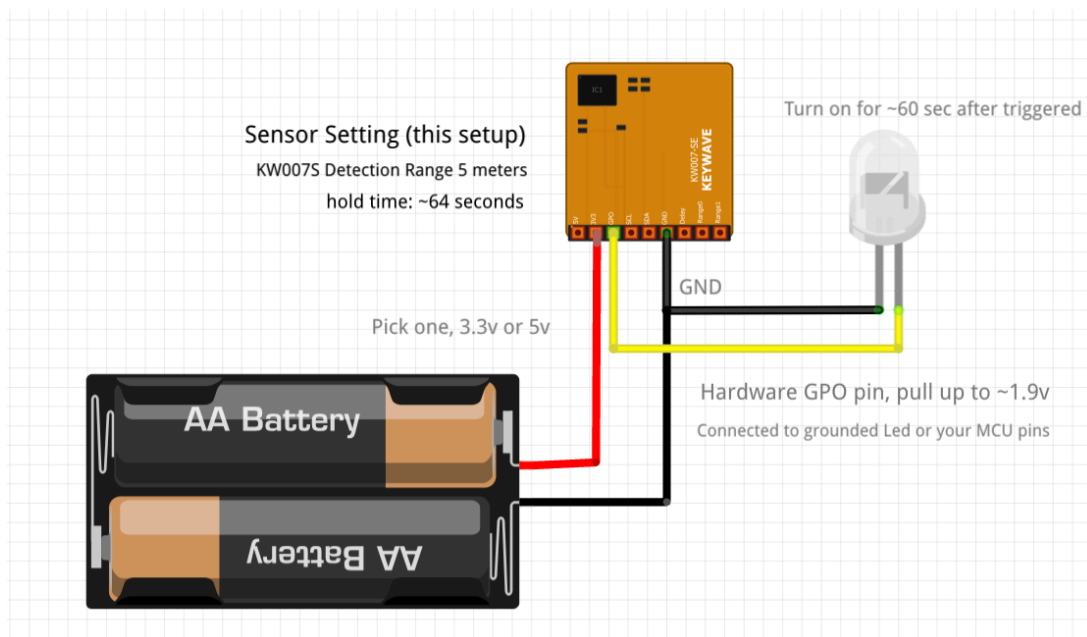
- Hold time: ~64 seconds

During the hold time, the output remains HIGH continuously, effectively indicating: “Motion detected” for the entire duration.

***No MCU or programming is required —***

***This works as a fully hardware-based solution.***

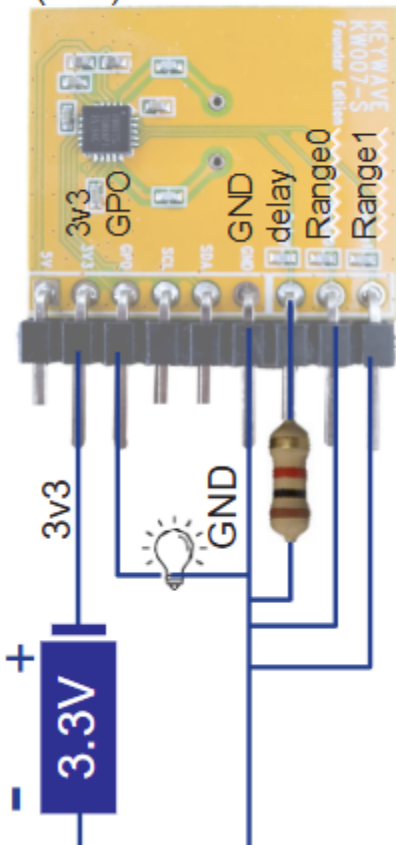
Any questions: chat, support or follow in X.com : <https://x.com/KeywaveT> or search our ID: @Keywave  
or Email: Q@keywavetech.co.uk



**KW007SE**

Range : Level 1

(~0.1m)

GPO Hold Time: 1 sec  
(Max)**Need a different range or timing?**

The KW007-SE allows configuration of **detection range** and **hold time** in two ways:

**Method 1) Hardware Configuration (Resistor-Based)**

Range and delay can be adjusted by connecting resistors to configuration pins.

Examples:

- **~10 cm detection range:**

Connect Range0 and Range1 directly to GND (no resistor)

- **~1 second hold time:**

Add a **270 kΩ resistor** between the Delay pin and GND

All configuration combinations are listed in the table below (datasheet section).

An **EVK with DIP switches** is available for quick evaluation and tuning.

You can find our EVK, sensors and other accessories here via DigiKey.com:

<https://www.digikey.com/en/supplier-centers/keywave-technology>

**Method 2) Software Configuration (I<sup>2</sup>C) with MCU**

For more flexibility, connect the **SDA/SCL (I<sup>2</sup>C)** pins to an MCU.

- Configure range, hold time, and other parameters via software
  - Full API and programming guide provided
  - Supports common platforms: ESP32, Arduino, Raspberry Pi, Pico, windows application etc.

If I<sup>2</sup>C is not used, the **GPO pin alone** is sufficient for motion-trigger output.

That's not all, with **MCU and I<sup>2</sup>C**, our sensor also can tell you motion is coming into the sensor or moving away from the sensor, it has different values and flags for your advanced application.

**Connected to MCU for Advanced Features**

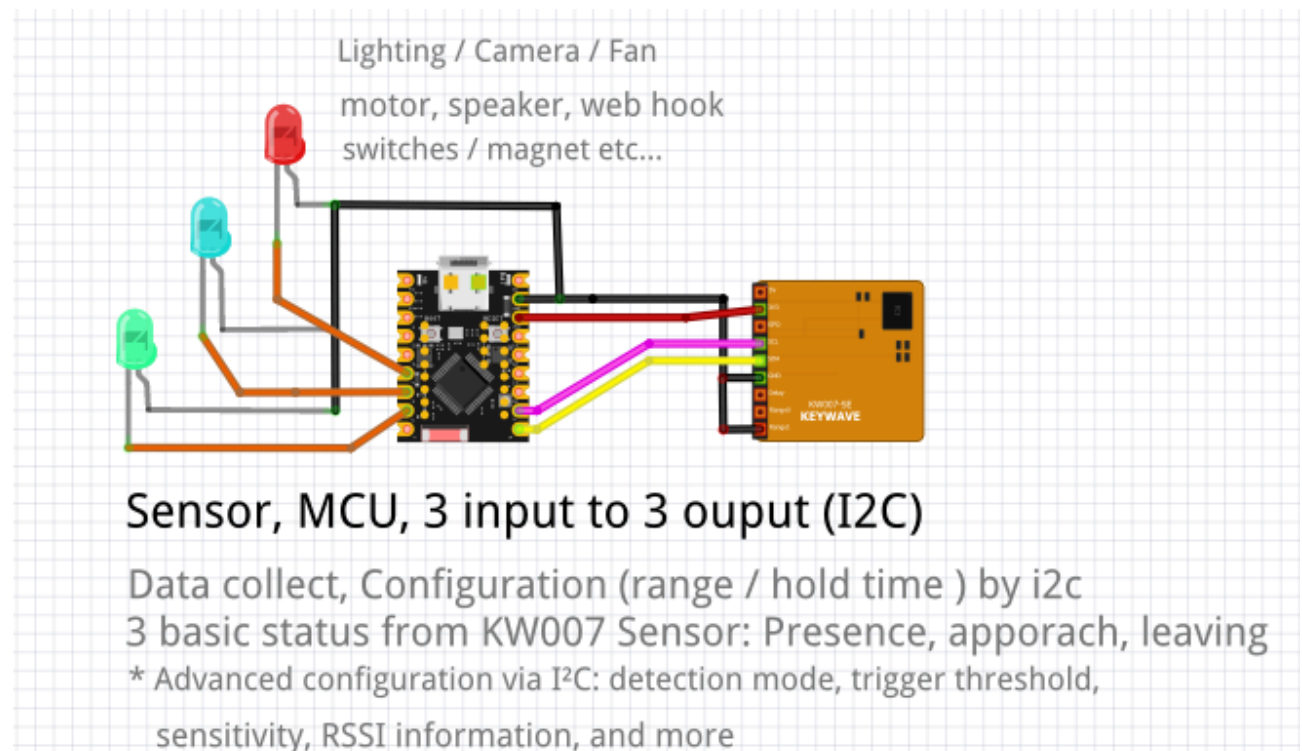
With MCU integration via **I<sup>2</sup>C**, the sensor provides more than just a trigger signal, it can adjust detection range, hardware GPO hold time, It also can indicate whether motion is **approaching or moving away** from the sensor, using dedicated values and flags for advanced applications such as directional detection and behavior analysis.

**Additional Features:**

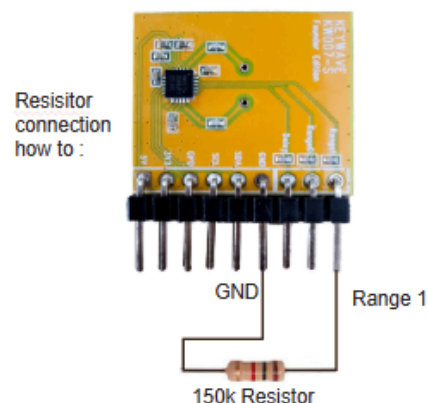
- **Through-material sensing:** Operates through non-metal materials, enabling concealed installation behind enclosures or walls
- **Ultra-low power consumption:**
  - ~30  $\mu$ A (KW007-SE)
  - ~100  $\mu$ A (KW007-L)
- **Stable and Reliable:** Operates alongside Wi-Fi and BLE without interference.
- Not affected by common environmental factors such as shadows, lighting changes, airflow, vibration, temperature variations, or sound.

**Available Models**

| Model    | Max Range |
|----------|-----------|
| KW007-SE | 5 m       |
| KW007-L  | 12 m      |



## Range and hold time by resistor Table



- Range setting

| Set Distance | Level 1 | Level 2 | Level 3 | Level 4 | Level 5 | Level 6 | Level 7 | Level 8 | *All Max |
|--------------|---------|---------|---------|---------|---------|---------|---------|---------|----------|
| Range 0      | 0       | 150kΩ   | 270kΩ   | *Open   | 0       | 150kΩ   | 270kΩ   | *Open   | *Open    |
| Range 1      | 0       | 0       | 0       | 0       | 150kΩ   | 150kΩ   | 150kΩ   | 150kΩ   | *Open    |
| KW007-S      | 0.1m    | 0.2m    | 0.3m    | 0.5m    | 0.7m    | 1m      | 1.4m    | 2m      | 2.0m     |
| KW007-L      | 5m      | 6m      | 7m      | 8m      | 9m      | 10m     | 11m     | 12m     | 12m      |
| KW007-SE     | 0.1m    | 0.2m    | 0.3m    | 0.5m    | 0.7m    | 1m      | 2m      | 4m      | 4m       |

\*0 = connect to GND

\* Open = no resistor fitted ( $\infty \Omega$ ) and not connect to GND

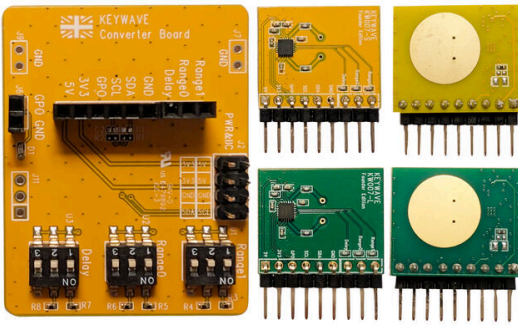
**\*Note on Detection Range:** Listed distances are maximum reference values. Since detection is based on reflection signal strength, the actual range may vary depending on environmental factors, object size, and the material or thickness of the product housing.

- Hold time setting

| Set Delay         | Max           | Setting 1     | Setting 2     | Setting 3  |
|-------------------|---------------|---------------|---------------|------------|
| Connect Resistor: | Resistor Open | Resistor 270k | Resistor 150K | Resistor 0 |
| KW007-S           | 64 sec.       | 1 sec.        | 2 sec.        | 4 sec      |
| KW007-L/SE        | 64 sec.       | 1 sec.        | 32 sec.       | 16 sec.    |

**\*Hold-Time Logic:** When motion is detected, the GPO output pulls high. The output remains high for the duration of the set hold time. If motion is detected again while the output is already high, the hold timer **restarts**, continuously extending the high-level duration.

We also provide EVK for faster hardware test:



A complete sensor set with an easy-to-use evaluation board for KW007.

DIP switches allow quick configuration of:

- Detection range
- Output signal hold time

Designed for rapid testing, tuning, and integration into your application.

---

## **KW007 Motion Sensor Module & EVK** **- Reliable Motion Detection Beyond PIR -**

Compact 5.8GHz Doppler motion sensor with adjustable range, low power operation, and simple GPIO integration.

---

### **KW007 ,what it is:**

- Compact 5.8GHz Doppler motion sensor module
  - Adjustable detection range (10 cm – 12 m depending on model)
  - < 30  $\mu$ A system current
  - Digital GPIO trigger output
  - Optional I<sup>2</sup>C programmable control
  - Example code provided
- 

**DIP Switch and detail Datasheet in Github:**

[https://github.com/namexavailable/Keywave\\_Modules/blob/main/KW007/KW007\\_Modules\\_EVK\\_Guide.pdf](https://github.com/namexavailable/Keywave_Modules/blob/main/KW007/KW007_Modules_EVK_Guide.pdf)

**Ai ready - Programming Guide:**

[https://github.com/namexavailable/Keywave\\_Modules/blob/main/KW007/KW007\\_Programming\\_Guide.pdf](https://github.com/namexavailable/Keywave_Modules/blob/main/KW007/KW007_Programming_Guide.pdf)

**Ai and maker ready - API and Code example plus explanation in comments:**

[https://github.com/namexavailable/Keywave\\_Modules/tree/main/KW007/code](https://github.com/namexavailable/Keywave_Modules/tree/main/KW007/code)

---

**Purchase Link in Digikey:**

<https://www.digikey.co.uk/en/products/detail/keywave-technology-limited/KWEVK0072SE2L2602/28960699>



## What It Does

- *Accurately and quickly detects active motion*
- *Outputs digital HIGH signal via GPO*
- *Configurable dynamic and provides directional data (toward / away) via I<sup>2</sup>C*
- *Operates reliably through plastic enclosures*
- *Stable against airflow, vibration, Wi-Fi and BLE interference*

## What It Is NOT

- Not a static human presence sensor  
(With MCU processing and sensitivity tuning, presence-like behavior may be approximated.)
  - Not a camera
  - Not a high-power radar module
  - Not heat-based detection (unlike PIR)
- 

## Compare with PIR:

Lower false triggers.

Works through plastic and other non-metal material.

Less sensitive to temperature and lighting.

## What range can I expect?

10 cm – 12 m (model dependent).

Performance depends on enclosure and environment.

## How do I evaluate it?

Use KW007 EVK.

No firmware required for basic testing.

Make something great with MCU via I2C

---

## Support and before purchase questions:

We will provide example implementations, code and guide via our website: [keywavetech.co.uk](http://keywavetech.co.uk)

If you need urgent support please mail to [w@keywavetech.co.uk](mailto:w@keywavetech.co.uk) or contact us via LinkedIn:

<https://www.linkedin.com/company/106465373>

Chat, support or follow in **X.com** : <https://x.com/KeywaveT> or search our ID: @KeywaveT

Please visit our website for most updated information: [www.keywavetech.co.uk](http://www.keywavetech.co.uk)

## **Applications, Integration & Performance Overview**

### **Typical Applications**

KW007 is designed for reliable motion-triggered activation in environments where conventional PIR sensors may suffer from false triggering or environmental instability.

### **Smart Building & Energy Applications**

- Smart lighting control
- HVAC activation
- Corridor and stairwell automation
- Energy-saving occupancy-based switching
- Lighting inside enclosed fixtures

### **Commercial & Public Installations**

- Interactive kiosks and displays
- Digital signage wake-on-motion
- Automatic switches and panels
- Equipment activation in public spaces
- Access control pre-trigger systems

### **Industrial & Embedded Systems**

- Machine wake-on-motion
- Enclosed equipment detection
- Temperature-variable environments
- Vibration-prone installations
- Compact embedded product designs



## System Integration Overview

KW007 is designed for flexible integration, from simple hardware-only trigger systems to advanced programmable platforms.

### Option A – Direct GPIO Integration (No MCU Required)

#### **KW007 → GPO Output → Relay / Logic Input**

- Digital HIGH output when motion detected
- Adjustable pulse duration (1–64 seconds)
- Ideal for direct switching applications
- Minimal development effort

### Option B – MCU + I<sup>2</sup>C Advanced Integration

#### **KW007 → I<sup>2</sup>C → MCU ( Cloud / IoT Platform) → Pin to your applications**

- Adjustable detection range
- Adjustable sensitivity
- Adjustable output duration
- Directional data (approaching / moving away)
- Real-time configuration control
- Event logging and condition-based programming

Example firmware is available to accelerate integration.

## Comparison: PIR vs KW007

KW007 is not a PIR replacement in all cases — it is designed for applications requiring greater stability and enclosure flexibility.

| Feature                      | PIR Sensor                | KW007                                             |
|------------------------------|---------------------------|---------------------------------------------------|
| Detection Principle          | Infrared heat change      | <b>RF Doppler motion</b>                          |
| Temperature Sensitivity      | High                      | <b>Low</b>                                        |
| False Trigger Susceptibility | Heat / airflow / sunlight | <b>Significantly reduced</b>                      |
| Detection Through Plastic    | No                        | <b>Yes (non-metal)</b>                            |
| Output Data                  | Trigger only              | <b>Trigger + Direction (I<sup>2</sup>C)</b>       |
| Environmental Stability      | Moderate                  | <b>High</b>                                       |
| Integration                  | Very simple               | <b>Simple (GPIO) or advanced (I<sup>2</sup>C)</b> |
| Power Consumption            | Very low                  | <b>Low (&lt;30 µA typical)</b>                    |

---

## Detection Characteristics

KW007 is designed to detect active motion.

At close range, small movements such as hand gestures can trigger detection.

At longer distances, larger displacement is required to maintain detection stability and prevent false triggering. Long-range performance depends on:

- Target size
- Target speed
- Mounting height
- Installation angle
- Enclosure material and thickness

Detection ranges listed in specifications represent maximum reference values under optimal conditions.

## Engineering-Ready Design

KW007 supports both rapid evaluation and scalable production deployment.

- Adjustable via hardware resistor configuration
- Fully programmable via I<sup>2</sup>C
- Example code provided
- Production pre-configuration available for volume orders
- Compact footprint (~22 mm class module)
- Designed for low-power systems

### Support and before purchase questions:

We will provide example implementations, code and guide via our website: [keywavetech.co.uk](https://keywavetech.co.uk)

If you need urgent support please mail to [w@keywavetech.co.uk](mailto:w@keywavetech.co.uk) or contact us via LinkedIn:

<https://www.linkedin.com/company/106465373>

Chat, support or follow in **X.com** : <https://x.com/KeywaveT> or search our ID: @KeywaveT

Please visit our website for most updated information: [www.keywavetech.co.uk](https://www.keywavetech.co.uk)